Compact-to-Hausdorff Continuous Bijection Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of compact-to-hausdorff continuous bijection theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

GT-02 / General Topology / Report v1

I. TARGET AND IMPORT CONTRACT

Theorem I.1. *If X is compact, Y is Hausdorff, and $f : X \to Y$ is a continuous bijection, then f is a homeomorphism.*

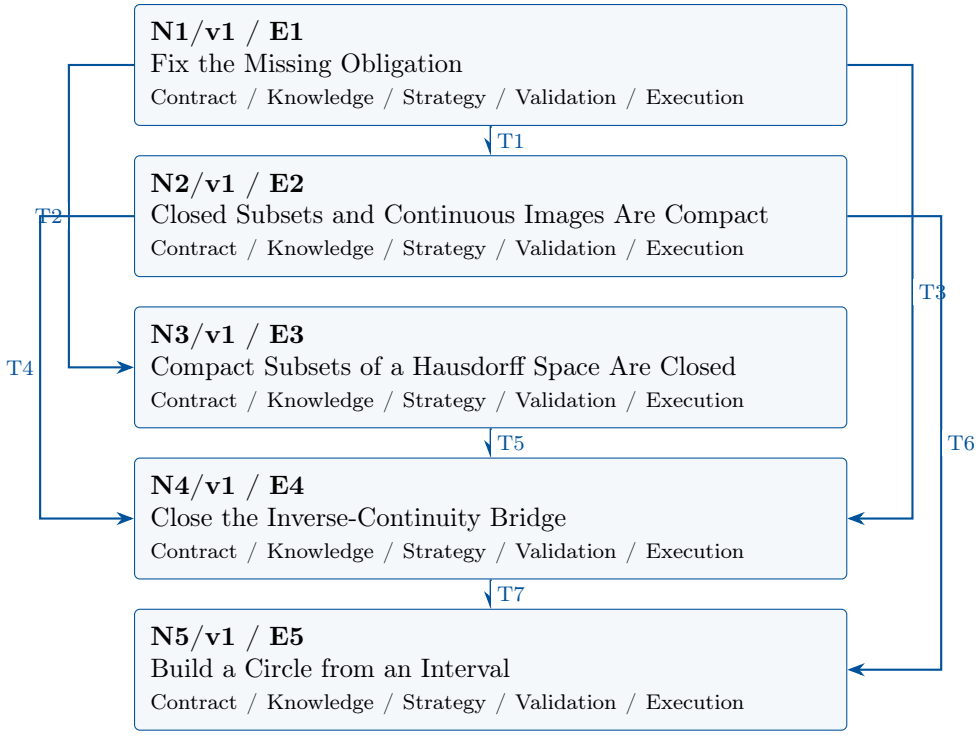
A homeomorphism is a bijection continuous in both directions. The task is therefore to prove continuity of the inverse. The source X need not initially be assumed Hausdorff; the conclusion makes it homeomorphic to a Hausdorff space.

A. Foundations and permitted imports

compactness means every open cover has a finite subcover. Hausdorff means distinct points have disjoint open neighborhoods. A closed map sends closed subsets to closed subsets. The proof below derives the supporting compactness lemmas directly from these definitions.

For the final application only, import compactness of the closed interval, the quotient universal property, and $e^{2\pi it} = e^{2\pi is}$ exactly when $t - s \in \mathbb{Z}$.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The inverse set-map $h = f^{-1} : Y \to X$ exists; it remains to prove its continuity.

Receiving activity. N2: Closed Subsets and Continuous Images Are Compact

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 and the open-cover definition of compactness.

T2: N1/E1/v1 to N3. The inverse set-map $h = f^{-1} : Y \to X$ exists; it remains to prove its continuity.

Receiving activity. N3: Compact Subsets of a Hausdorff Space Are Closed

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1 supplies the Hausdorff target; take an arbitrary compact subset $K \subset Y$.

T3: N1/E1/v1 to N4. The inverse set-map $h = f^{-1} : Y \to X$ exists; it remains to prove its continuity.

Receiving activity. N4: Close the Inverse-Continuity Bridge

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and N3/E3.

T4: N2/E2/v1 to N4. For every closed $C \subset X$, the image $f(C)$ is compact.

Receiving activity. N4: Close the Inverse-Continuity Bridge

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and N3/E3.

T5: N3/E3/v1 to N4. Every compact subset of a Hausdorff space is closed.

Receiving activity. N4: Close the Inverse-Continuity Bridge

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N1/E1, N2/E2, and N3/E3.

T6: N2/E2/v1 to N5. For every closed $C \subset X$, the image $f(C)$ is compact.

Receiving activity. N5: Build a Circle from an Interval

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N4/E4, compactness of $[0, 1]$, the quotient topology, and the exponential circle parametrization.

T7: N4/E4/v1 to N5. f is a homeomorphism.

Receiving activity. N5: Build a Circle from an Interval

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: N2/E2, N4/E4, compactness of $[0, 1]$, the quotient topology, and the exponential circle parametrization.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Fix the Missing Obligation

Object and role. Fix the Missing Obligation. Establish the following local result: The inverse set-map $h = f^{-1} : Y \to X$ exists; it remains to prove its continuity.

Inputs. X compact, Y Hausdorff, and $f : X \to Y$ a continuous bijection.

C — Contract

Own E1/v1: The inverse set-map $h = f^{-1} : Y \to X$ exists; it remains to prove its continuity.

K — Knowledge

Bijection and topology definitions

S — Strategy

Target inverse continuity

V — Validation

Check each hypothesis separately

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Check compactness, Hausdorffness, continuity, and bijectivity separately.

N1.2 Define the inverse as a set-map.

N1.3 Reduce inverse continuity to closed inverse images.

Export E1/v1. The inverse set-map $h = f^{-1} : Y \to X$ exists; it remains to prove its continuity.

Acceptance gate. Check each hypothesis separately Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Check each hypothesis separately Recheck dependent nodes N2, N3, N4, N5.

B. N2/v1: Closed Subsets and Continuous Images Are Compact

Object and role. Closed Subsets and Continuous Images Are Compact. Establish the following local result: For every closed $C \subset X$, the image $f(C)$ is compact.

Inputs. N1/E1 and the open-cover definition of compactness.

C — Contract

Own E2/v1: For every closed $C \subset X$, the image $f(C)$ is compact.

K — Knowledge

Open covers and preimages

S — Strategy

Transfer finite-subcover property

V — Validation

Use closed source sets

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Add the open complement of C to a cover and use compactness of X.

N2.2 Pull an open cover of $f(C)$ back along the restricted continuous map.

N2.3 Push the resulting finite subcover forward.

Export E2/v1. For every closed $C \subset X$, the image $f(C)$ is compact.

Acceptance gate. Use closed source sets Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: Use closed source sets Recheck dependent nodes N4, N5.

C. N3/v1: Compact Subsets of a Hausdorff Space Are Closed

Object and role. Compact Subsets of a Hausdorff Space Are Closed. Establish the following local result: Every compact subset of a Hausdorff space is closed.

Inputs. N1/E1 supplies the Hausdorff target; take an arbitrary compact subset $K \subset Y$.

C — Contract

Own E3/v1: Every compact subset of a Hausdorff space is closed.

K — Knowledge

Disjoint neighborhoods

S — Strategy

Separate outsider from compact set

V — Validation

Use finite intersections after a finite subcover

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Separate an outsider from each point of K by disjoint open neighborhoods.

N3.2 Use compactness to select finitely many neighborhoods covering K.

N3.3 Intersect the finitely many outsider neighborhoods to prove the complement open.

Export E3/v1. Every compact subset of a Hausdorff space is closed.

Acceptance gate. Use finite intersections after a finite subcover Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: Use finite intersections after a finite subcover Recheck dependent nodes N4, N5.

D. N4/v1: Close the Inverse-Continuity Bridge

Object and role. Close the Inverse-Continuity Bridge. Establish the following local result: f is a homeomorphism.

Inputs. N1/E1, N2/E2, and N3/E3.

C — Contract

Own E4/v1: f is a homeomorphism.

K — Knowledge

Closed-set continuity criterion

S — Strategy

Recognize f as closed map

V — Validation

Use bijection to identify preimages

E — Execution

Carry out the numbered deductions and publish E4/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N4.1 Apply the compact-subset lemma to $f(C)$ for each closed C.

N4.2 Use $h^{-1}(C) = f(C)$ to show the inverse is continuous.

N4.3 Combine both continuous inverse maps.

Export E4/v1. f is a homeomorphism.

Acceptance gate. Use bijection to identify preimages Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N4: Use bijection to identify preimages Recheck dependent nodes N5.

E. N5/v1: Build a Circle from an Interval

Object and role. Build a Circle from an Interval. Establish the following local result: $[0,1]/(0 \sim 1)$ is homeomorphic to S^1 by $[t] \mapsto e^{2\pi it}$.

Inputs. N2/E2, N4/E4, compactness of $[0,1]$, the quotient topology, and the exponential circle parametrization.

C — Contract

Own E5/v1: $[0,1]/(0 \sim 1)$ is homeomorphic to S^1 by $[t] \mapsto e^{2\pi it}$.

K — Knowledge

Quotient universal property

S — Strategy

Apply theorem to compact quotient

V — Validation

Check endpoint equivalence exactly

E — Execution

Carry out the numbered deductions and publish E5/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N5.1 Descend the continuous endpoint-identifying map through the quotient.

N5.2 Check exactly the endpoint pair is identified and the map is onto.

N5.3 The quotient is compact and the circle Hausdorff, so the already proved theorem applies.

Export E5/v1. $[0,1]/(0 \sim 1)$ is homeomorphic to S^1 by $[t] \mapsto e^{2\pi it}$.

Acceptance gate. Check endpoint equivalence exactly Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N5: Check endpoint equivalence exactly Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Fix the Missing Obligation

Bijectivity defines a set-map $h = f^{(-1)} : Y \to X$. Its continuity is not assumed. We will show that for every closed C in X, $h^{(-1)}(C)$ is closed in Y. This closed-set criterion is equivalent to continuity by taking complements.

B. N2 — Closed Subsets and Continuous Images Are Compact

Let C be closed in compact X. Any cover of C by relatively open sets can be represented by sets open in X. Add the open complement $X \setminus C$. These sets cover X, so finitely many suffice. Removing the complement leaves a finite subcover of C. Thus C is compact.

Now let K be compact and $g : K \to Z$ continuous. An open cover of $g(K)$ pulls back under g to an open cover of K. Choose finitely many preimages covering K; their original sets cover $g(K)$. Hence continuous images of compact sets are compact.

Apply these two lemmas to f restricted to C. The result is that $f(C)$ is compact in Y.

C. N3 — Compact Subsets of a Hausdorff Space Are Closed

Let K be compact in Hausdorff Y and choose y outside K. For every x in K choose disjoint open neighborhoods U_x of x and V_x of y. Finitely many U_x, say $U_{x_1}, \ldots, U_{x_m}$, cover K. The intersection $V = \bigcap_{j=1}^{m} V_{x_j}$ is an open neighborhood of y disjoint from their union and thus from K.

Every point outside K has such a neighborhood, so $Y \setminus K$ is open and K is closed. If K is empty, closedness is immediate. Finiteness is essential: an arbitrary intersection of open neighborhoods need not be open.

D. N4 — Close the Inverse-Continuity Bridge

For every closed C in X, N2 proves $f(C)$ compact and N3 proves it closed in Y. Therefore f is a closed map. Since h is the inverse bijection,

$$h^{-1}(C) = f(C).$$

This is closed for every closed C, so h is continuous. The original f was continuous, and thus is a homeomorphism. □

E. N5 — Build a Circle from an Interval

Let Q be the quotient of [0,1] identifying only 0 with 1, and let $q : [0, 1] \to Q$ be the quotient map. Define

$$g : Q \to S^1, \qquad g([t]) = e^{2\pi i t}.$$

It is well-defined because the endpoints have the same value. It is continuous by the quotient topology, since g composed with q is continuous. It is onto, and it is one-to-one because equal exponentials imply $t - s$ is an integer; on [0,1] this gives either $t = s$ or the identified endpoint pair.

The interval is compact, q is continuous and onto, so Q is compact by N2's lemma. The circle is Hausdorff as a subspace of R^2. N4 therefore proves g is a homeomorphism without needing to analyze inverse continuity separately at the glued endpoint.

□

V. WORKED ORCHESTRATION

Let Q be the quotient of [0,1] identifying only 0 with 1, and let $q : [0, 1] \to Q$ be the quotient map. Define

$$g : Q \to S^1, \qquad g([t]) = e^{2\pi i t}.$$

It is well-defined because the endpoints have the same value. It is continuous by the quotient topology, since g composed with q is continuous. It is onto, and it is one-to-one because equal exponentials imply $t - s$ is an integer; on [0,1] this gives either $t = s$ or the identified endpoint pair.

The interval is compact, q is continuous and onto, so Q is compact by N2's lemma. The circle is Hausdorff as a subspace of R^2. N4 therefore proves g is a homeomorphism without needing to analyze inverse continuity separately at the glued endpoint.

The application consumes the already proved N4 theorem. It supplies no premise back to N4.

A. Boundary and hypothesis audit

Hausdorffness cannot be dropped: the identity from [0,1] with its usual topology to the same set with the indiscrete topology is a continuous bijection from a compact source, but its inverse is not continuous. Bijectivity also matters: a continuous surjection may identify distinct points and need not have an inverse at all.

If both spaces are empty the conclusion is immediate. Finite intersections, rather than arbitrary intersections, are the decisive use of compactness in N3.

Export the closed-subset compactness lemma, continuous-image compactness lemma, Hausdorff compact-set closedness lemma, and the continuous-bijection criterion. The circle application supplies a concrete quotient-space example for later algebraic topology constructions.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

Consider the continuous bijection $f : [0, 2\pi) \to S^1$, $f(t) = e^{it}$. The source is not compact. Points $e^{i(2\pi - 1/n)}$ approach 1, but their inverse values approach 2π rather than $f^{(-1)}(1) = 0$. Thus the inverse is discontinuous.

- **Node:** N1
- **State:** STALLED
- **Failing component:** Validation
- **Failed hypothesis:** compact source
- **Observed:** half-open angle interval
- **Repair target:** source topology/model
- **Blocked export:** homeomorphism

Knowledge recalls the compactness requirement. Validation isolates the cut at angle zero. Strategy replaces the half-open representative model by the compact endpoint quotient. Execution verifies the quotient map's continuity and bijectivity. Contract publishes the corrected homeomorphism and preserves the original counterexample as an explanation of why the repair was needed.

B. Hypothesis boundary

Hausdorffness cannot be dropped: the identity from [0,1] with its usual topology to the same set with the indiscrete topology is a continuous bijection from a compact source, but its inverse is not continuous. Bijectivity also matters: a continuous surjection may identify distinct points and need not have an inverse at all.

If both spaces are empty the conclusion is immediate. Finite intersections, rather than arbitrary intersections, are the decisive use of compactness in N3.

Export the closed-subset compactness lemma, continuous-image compactness lemma, Hausdorff compact-set closedness lemma, and the continuous-bijection criterion. The circle application supplies a concrete quotient-space example for later algebraic topology constructions.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. MIT Introduction to Topology: compactness and separation material.